

within original shape

B1

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- (a) (i) at 22.5 °C, $R_T = 1600\Omega$ or 1.6 k Ω
total resistance = 800 Ω

C1

A1 [2]

- (ii) *either* use of potential divider formula *or* current = 9 / 2000 (4.5 mA)
 $V = (0.8/2.0) \times 9$ $V = (9/2000) \times 800$
 $= 3.6V$ $= 3.6V$

C1

A1 [2]

- (b) (i) total resistance = $4/5 \times 1200$
 $= 960\Omega$

C1

A1 [2]

- (ii) for parallel combination, $1/960 = 1/1600 + 1/R_T$
 $R_T = 2400\Omega$ / 2.4 k Ω
temperature = 11 °C

C1

A1 [2]

- (c) e.g. only small part of scale used / small sensitivity
non-linear

B1

B1 [2]

(any two sensible suggestions, 1 each, max 2)

(e.g. from 0 °C to 25 °C, the voltmeter reading varies from about 4.3 V to 3.6 V so
only small part of the voltmeter scale is used / small sensitivity in V per °C change)

[10]

Comments: Most answered this question well. In part (b) (ii), lack of care saw a few
misread the temperature from Fig. 5.1.

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- (a) (i) Resistance of the bulb is defined as the ratio $\frac{V}{I}$ where V is the potential